Linear Algebra

Matrix

Oct.27th - Oct.29th, 2026 Lecture

Basic calculation of matrix

Before addressing “Inverse matrix”, let’s first go over the most fundamental matrix operations. Some points were explained in previous notes, while others are described here specifically for the purpose of calculating the inverse matrix.

Matrices – Basic Calculation

1. Scalar multiplication of a matrix is performed by multiplying each of its entries by the scalar.

$$k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

2. Matrix addition is a component-wise (or elementwise) operation. It is only defined when two matrices have the exact same dimensions.

$$A + B = C$$

3. Matrix multiplication (AB) is not elementwise. It is defined based on the inner product (dot product) of rows and columns.

$$A \times B = C$$

4. Matrix multiplication is directional, and the commutative law does not hold. Thus,

$$ABC \neq BCA \neq CAB$$

Inverse matrix

An **inverse matrix** represents the multiplicative inverse of a square matrix and was devised to perform the equivalent of division—one of the four basic arithmetic operations—within the context of matrices.
To solve **system of linear equations** below:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = s_1 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = s_n \end{cases}$$

By stripping away the variables and the equal signs, we can rewrite this system as a single matrix equation:

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} s_1 \\ \vdots \\ s_n \end{bmatrix}$$

And simply, matrix equations above also expressed as below:

$$AX = b$$

When vector X is the pairs solution for equation above, it can be evaluated via move A along to right side. Then,

$$X = \frac{b}{A}$$

In general, a matrix cannot appear in a denominator; in other words, the multiplicative inverse of a matrix is not generally defined. The concept of an inverse matrix is introduced to address this within the context of matrix multiplication.

In the system of matrix, Identity matrix($\boldsymbol{I}$) act as a number 1. If it is true that the inverse matrix replaces the role of division, Identity matrix will be obtained via multiply matrix and its invers matrix as like below:

$$AA^{-1} = I$$

Instantly, matrix itself can be inverse matrix of its inverse matrix(fah). In this case, **commutative law holds**.

Matrices – Inverse matrix

Inverse matrixes are performing the following operations by multiplying their original matrix:

$$AA^{-1} = A^{-1}A = I$$

Example 1:
Prove that the statement below is true:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Sol)

$$let\ inverse\ function\ of\ matrix\ AB\ to\ X$$

$$X\ suppose\ to\ be\ satisfied\ X(AB) = (AB)X = I.$$

$$While\ X = B^{-1}A^{-1}:$$

$$B^{-1}A^{-1}AB = B^{-1}IB = B^{-1}B = I$$

$$ABB^{-1}A^{-1} = AIA^{-1} = AA^{-1} = I$$

Then how to find Inverse matrix? For a 2x2 square matrix, the inverse can be found quite easily. Let us consider arbitrary matrices A and B. Then, multifaction of two matrices:

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x & y \\ z & r \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} ax+bz & ay+br \\ cx+dz & cy+dr \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$ax + bz = 1 \cdots \mathbb{a}$$

$$cy + dr = 1 \cdots \mathbb{b}$$

$$ay + br = 0 \cdots \mathbb{c}$$

$$cx + dz = 0 \cdots \mathbb{d}$$

$$From\ \mathbb{c}, \mathbb{d} \rightarrow\ x = -\frac{dz}{c}, y = -\frac{br}{a}, z = -\frac{cx}{d}, r = -\frac{ay}{b}$$

$$Substitute\ all\ of\ those\ to\ \mathbb{a}, \mathbb{b}$$

$$-\frac{dz}{c}a + bz = 1 \rightarrow\ z = \frac{1}{\left(b - \frac{da}{c}\right)} = -\frac{c}{\det A}$$

$$ax - \frac{cx}{d}b = 1 \rightarrow\ x = \frac{1}{\left(a - \frac{cb}{d}\right)} = \frac{d}{\det A}$$

$$-\frac{br}{a}c + dr = 1 \rightarrow\ r = \frac{1}{\left(d - \frac{bc}{a}\right)} = \frac{a}{\det A}$$

$$cy - \frac{ay}{b}d = 1 \rightarrow\ y = \frac{1}{\left(c - \frac{da}{b}\right)} = -\frac{b}{\det A}$$

$$Thus, \quad \begin{bmatrix} x & y \\ z & r \end{bmatrix} = \begin{bmatrix} \frac{d}{\det A} & -\frac{b}{\det A} \\ -\frac{c}{\det A} & \frac{a}{\det A} \end{bmatrix} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Matrices – Inverse matrix

Since $\det A \neq 0$, Inverse matrix of square matrix A can be expressed as below:

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

And define the Adjoint Matrix $Adj(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Since determinant is in denominators, Inverse matrix cannot be defined when determinant is zero. Therefore, only matrices with a non-zero determinant have an inverse matrix.

So, how many inverse matrices can a single matrix have? The answer to this can be determined by the concept of the "uniqueness of the inverse matrix," which is described in the example and solution below.

Example 2:

Prove that the Uniqueness of the Inverse Matrix.

Sol)

$\forall A \in \{M \in M_{n\times n}(\mathbb{R}) | \det A \neq 0\}$,

$$Suppose\ \exists B, C\ s.t.\ BA = AB\ = I\ and\ CA = AC = I\ with\ B \neq C$$

$$ABC = BAC\ = C$$

$$ABC = BAC \rightarrow IC = BI$$

$$\Longrightarrow\ B = C\ (\because by\ calculation)$$

$$Since\ B \neq C\ by\ hyp.\ we\ have\ B = C\ \Longrightarrow \perp$$

$$\therefore Inverse\ matrix\ is\ unique$$

$$Meaning: For\ every\ n \times n\ matrixes\ A, assume\ two\ distinct\ inverse\ matrices\ B, C\ .$$

$$By\ the\ calculation, B\ equals\ to\ C, contradicts\ to\ the\ statement.$$

$$Thus, Inverse\ matrix\ is\ unique$$

Augmented matrix

In the first lecture and the accompanying lecture notes, we touched upon the concept of an augmented matrix very briefly. For a system of linear equations with n variables:

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} s_1 \\ \vdots \\ s_n \end{bmatrix}$$

The augmented matrix corresponding to the $n \times n$ square matrix is as follows:

$$\left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & s_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & s_n \end{array}\right]$$

In this case, the following three operations can be applied to this system of linear equations (i.e., its augmented matrix).

Techniques- elementary row operation

1. Row substitution

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n+1} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{i,n+1} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{j,n+1} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k,n+1} \end{bmatrix} \xrightarrow{R_i \Rightarrow R_j} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n+1} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{j,n+1} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{i,n+1} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k,n+1} \end{bmatrix}$$

2. Row constant product

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n+1} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{i,n+1} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{j,n+1} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k,n+1} \end{bmatrix} \xrightarrow{cR_i \Rightarrow R_i} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n+1} \\ \vdots & \vdots & & \vdots \\ ca_{i1} & ca_{i2} & \cdots & ca_{i,n+1} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{j,n+1} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k,n+1} \end{bmatrix}$$

$$c \in \mathbb{K} \backslash \{0\}$$

3. Sum of rows

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n+1} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{i,n+1} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{j,n+1} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k,n+1} \end{bmatrix} \xrightarrow{cR_i + R_j \Rightarrow R_j} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1,n+1} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{i,n+1} \\ \vdots & \vdots & & \vdots \\ ca_{i1} + a_{j1} & ca_{i2} + a_{j2} & \cdots & ca_{i,n+1} + a_{j,n+1} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k,n+1} \end{bmatrix}$$

$$c \in \mathbb{K} \backslash \{0\}$$

We can convert an augmented matrix into a "row echelon matrix" through the following three operations.

A row echelon matrix is a matrix in which the elements are roughly in the form of an echelon on top and zero on the bottom. For example,

$$\begin{bmatrix} 1 & a & b & c & d \\ 0 & 0 & 2 & e & f \\ 0 & 0 & 0 & 1 & g \end{bmatrix}$$

If the leading entry(also called by “Pivot”) in each non-zero row is 1 like as below, defined as Reduced Row Echelon Form (RREF).

$$\begin{bmatrix} 1 & a & b & c & d \\ 0 & 0 & 1 & e & f \\ 0 & 0 & 0 & 1 & g \end{bmatrix}$$

Finally, the algorithm that transforms an m × n matrix M into a row echelon matrix by applying elementary row operations is called **Gaussian elimination**.

In particular, the operation that ultimately results in a reduced row echelon matrix is called **Gauss-Jordan elimination**.

When applying the Gauss-Jordan elimination method, **if any single row consists entirely of zeros**, the system of linear equations **has infinitely many solutions**. Alternatively, if the method **cannot be successfully applied**, it is concluded that the **system of linear equations has no solution**.

Example 3:

Solve instantaneous equation below via Gaussian elimination.

$$\begin{cases} x + y - z = 2 \\ 2x - y + 3z = 11 \\ x + 2y + z = 7 \end{cases}$$

Hint: To reduce calculation errors, it is best to perform calculations in a way that ensures a zero appears first in the bottom row.

Sol)

$$For\ its\ augmented\ matrix \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -1 & 3 & 11 \\ 1 & 2 & 1 & 7 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -1 & 3 & 11 \\ 1 & 2 & 1 & 7 \end{array}\right] \xrightarrow[R_2 - 2R_1 \Rightarrow R_2]{R_3 - R_1 \Rightarrow R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -3 & 5 & 7 \\ 0 & 1 & 2 & 5 \end{array}\right] \xrightarrow[R_3 + 3R_2 \Rightarrow R_3]{R_3 \Leftrightarrow R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 11 & 22 \end{array}\right] \xrightarrow{\frac{1}{11} R_3 \Leftrightarrow R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$Thus, z = 2, y = 1, x = 3$$

Example 4:

Solve instantaneous equation below via Gauss-Jordan elimination.

$$\begin{cases} x + y - z = 2 \\ 2x - y + 3z = 11 \\ x + 2y + z = 7 \end{cases}$$

Hint: To reduce calculation errors, it is best to perform calculations in a way that ensures a zero appears first in the bottom row.

Sol)

$$For\ its\ augmented\ matrix \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -1 & 3 & 11 \\ 1 & 2 & 1 & 7 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -1 & 3 & 11 \\ 1 & 2 & 1 & 7 \end{array}\right] \xrightarrow[R_2 - 2R_1 \Rightarrow R_2]{R_3 - R_1 \Rightarrow R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -3 & 5 & 7 \\ 0 & 1 & 2 & 5 \end{array}\right] \xrightarrow[R_3 + 3R_2 \Rightarrow R_3]{R_3 \Leftrightarrow R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 11 & 22 \end{array}\right] \xrightarrow[\frac{1}{11}R_3 \Rightarrow R_3]{}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow[R_1 - R_2 + R_3 \Rightarrow R_1]{R_2 - R_3 \Rightarrow R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$Thus, z = 2, y = 1, x = 3$$